DeepPrep-Based Spectral and Temporal Analysis of BOLD Signal Dynamics in Healthy Adults

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Problem / Motivation / Research Gap

- Traditional fMRI analysis focuses on spatial activation maps and overlooks temporal + spectral dynamics of BOLD signals
- Motion artifacts and preprocessing variability reduce reproducibility in neuroimaging studies
- Need for a scalable, reproducible pipeline to extract high-dimensional BOLD features in HPC environments

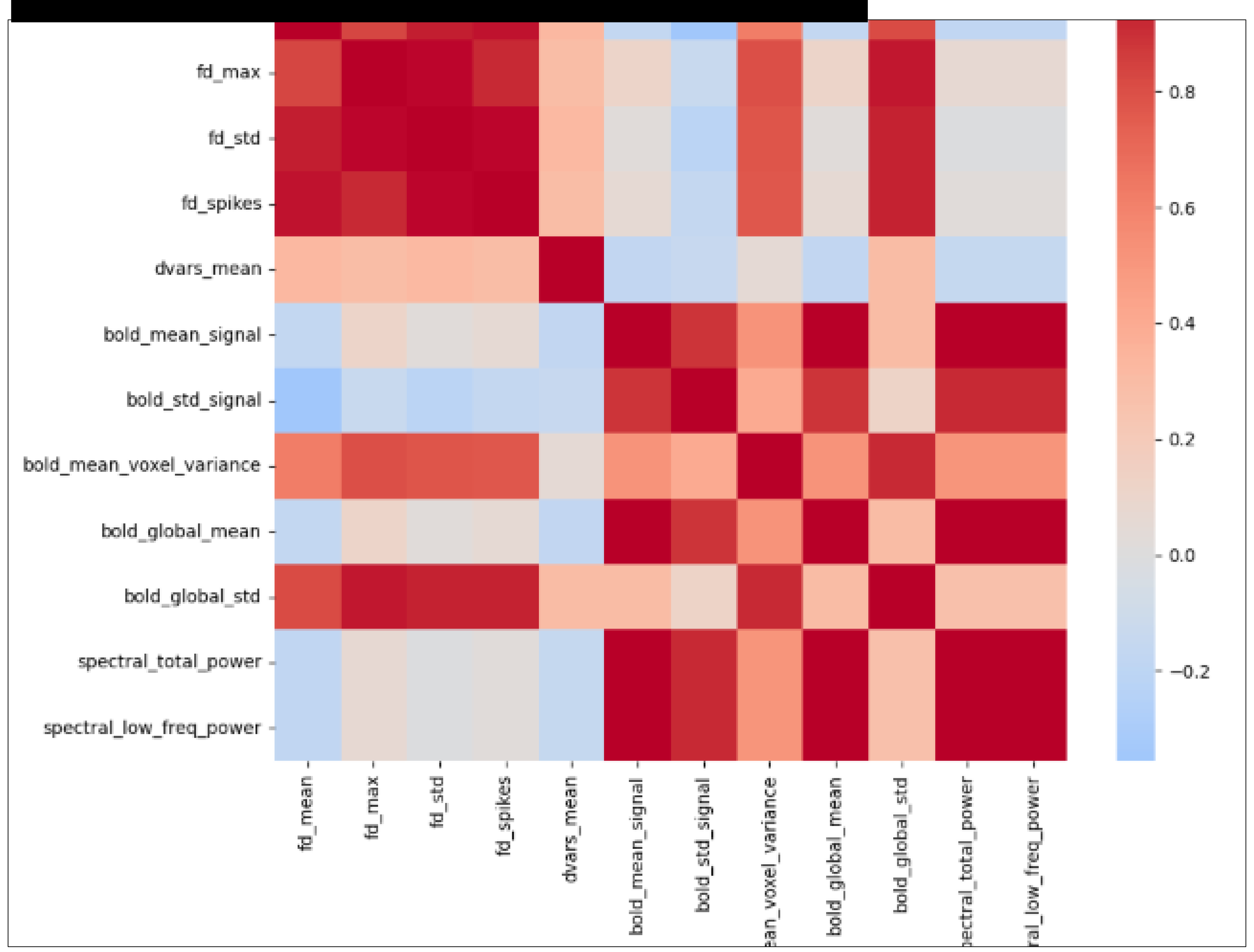
Methodology

- DeepPrep pipeline deployed on FABRIC HPC infrastructure
- BIDS-compliant OpenNeuro dataset (healthy adult cohort, n = 9)
- Preprocessing: alignment, normalization, and QC
- Feature extraction: Framewise Displacement (FD), DVARS, FC matrices, ALFF, spectral power metrics
- Containerized workflow (Docker) + Python analysis (Nilearn, NumPy, Pandas)

Implementation

- FABRIC compute environment (Ubuntu, 8-core, 32GB RAM)
- Automated DeepPrep execution pipeline
- HTML QC reports + workflow validation logs
- Post-processing scripts for motion + spectral feature extraction
- Correlation + regression analysis of extracted features

Feature Correlation Matrix



Results

- Low FD and DVARS indicate stable, low-motion cohort
- Strong ALFF dominance (>99% low-frequency power) confirms expected resting-state physiology
- Spectral features are largely independent of motion artifacts
- Correlation structure shows clear separation between:
  - motion metrics cluster
  - spectral features cluster
- Regression confirms no systematic motion-driven inflation of signal variance

Discussion / Conclusions / Future Work

- Pipeline successfully separates neural signal from motion-induced noise
- Spectral features provide robust representation of resting-state brain activity
- FABRIC + DeepPrep enables scalable, reproducible neuroimaging workflows
- Future work: apply pipeline to nicotine-dependent cohort and evaluate spectral shifts (ALFF/fALFF)
- Extend analysis with ML models for clinical prediction

Team Contributions

- **Scott Harris** – Power analysis, sample size estimation, statistical support, ran 3 subjects in pipeline, co-authored report
- **Caroline Lewis** – Pipeline design, integration of DeepPrep workflow, mock presentation, demo recording, final submission, ran 3 subjects in pipeline
- **Shokhsanam Uktamova** – Wrote 2-Pager proposal, ran 3 subjects in pipeline, co-authored report